

Figure F10 — Head-to-Head Tool Comparison

Same Liu 2023 cold-seep subset (200 MAG × 30 sample) on identical hardware

	EnvMeta	Krona [22]	MicrobiomeAnalyst [34]	Anvi'o [12]
PCoA on β -diversity	✓ 0.4 s	n/a	✓ ~10 s (web)	n/a
KO heatmap	✓ 0.5 s	n/a	partial	partial
Hierarchical HTML	✓ stackplot proxy	✓ TBD	✓ sunburst	n/a
Element cycle inference	✓ 1.0-18 s	✗	✗	✗
YAML hypothesis scoring	✓	✗	✗	✗
Standalone offline HTML SI	✓ 400 KB	partial	✗	partial

Note: Anvi'o benchmark not measured due to input-format mismatch (requires contigs.db + profile.db); MicrobiomeAnalyst row pending Web execution

Yellow = unique to EnvMeta; ✗ = capability not provided by tool